



FRC Systems

12x - Under The Bumper





Presentation Goals

- ☐ You will learn about the definition of an under the bumper
- ☐ You will understand how an under the bumper intake works
- ☐ You will learn about the advantages of an under the bumper
- ☐ You will understand how to model geometries and parts for the intake



Presentation Order

- ☐ Under the bumper - definition and differences
- ☐ Usage limitations and advantages
- ☐ Ground rules for design



Definition and differences

Core design principles

Difference in usage



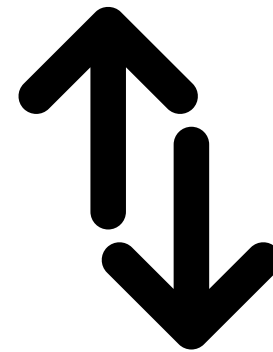
Difference in constraints

Difference in path



Usage limitations and advantages

Mechanical implications and in game difference



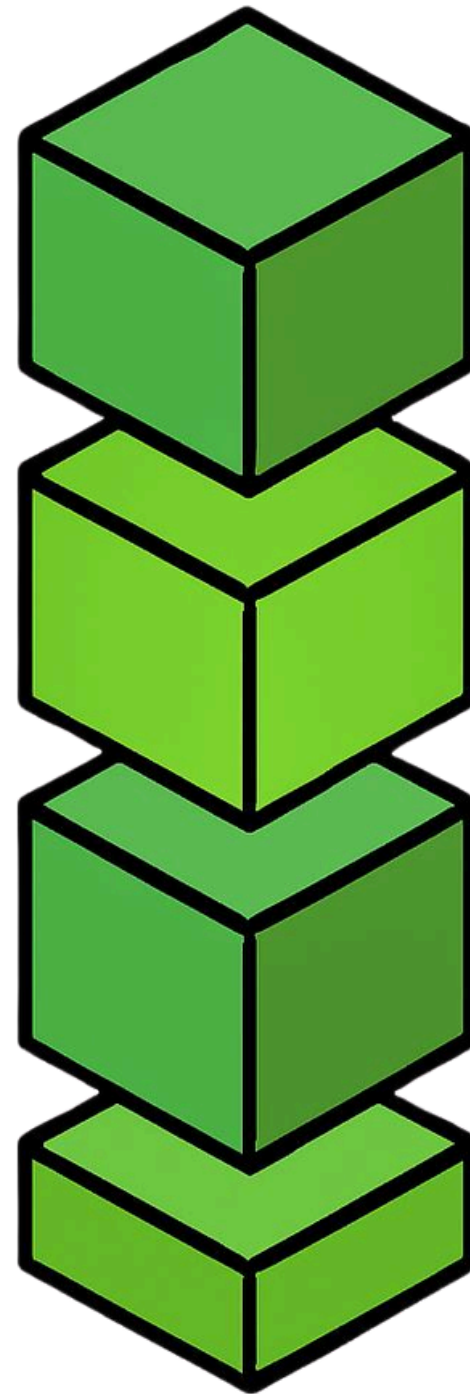
Difficulty in design vs Simplicity in design



Grouping rules for design

**Index is not complementary
– it is a must**

Always prototype



Easy to assemble and replace

“Keep it stupid simple”



Summary and questions